Audio amplifier based on transistors

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Abstract—This document presents the design, construction, and simulation of an audio amplifier based on transistors using a voltage divider configuration. The circuit incorporates an NPN 2N2222A transistor and a TIP32C transistor to amplify low-power audio signals intended for output devices such as speakers. The methodology includes detailed calculations and procedures to achieve an efficient design. Although the audio signals were amplified satisfactorily, the prototype reveals areas for improvement, such as thermal management, reduction of distortion, and precision in component selection, suggesting future optimizations for better performance and reliability. Among the topics addressed are transistor biasing, frequency response, and practical challenges in implementing electronic circuits.

Index Terms—Audio amplifier, transistors, divider of voltage, electronics, transistor biasing, frequency, speakers, capacitors, resistors.

I. INTRODUCTION

A transistor-based audio amplifier is an electronic circuit designed specifically to increase the power of the audio signals it receives, ensuring playback on output devices or speakers, allowing the audio to have higher volume and clarity. As the name suggests, these devices are based on the principle of amplification, where a transistor controls a larger current according to a much weaker input signal [1], [2]. This same principle often has different applications in areas of technology such as sound systems, musical instrumentation, cell phone sound, among others.

One of the main characteristics of an amplifier is its great ability to process low-power signals and convert them into stronger signals that can be controlled via speakers. This is achieved by using transistors, allowing multiple configurations, in which highlight the common-emitter, common-collector and push-pull configurations. Each of these configurations has specific characteristics that make them suitable for different amplification stages [3], [4].

A. Principle of operation of a transistor

The transistor falls into the category of being a device semiconductor that has three terminals; emitter, base and collector [1]. In Fig. 1 the visual diagram of bipolar transistors is observed.

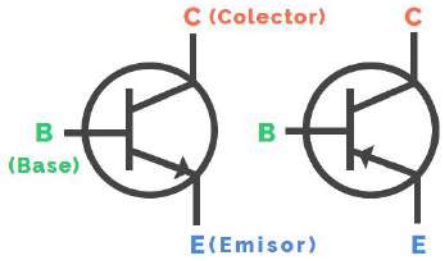


Fig. 1: Diagram of bipolar transistor symbols: NPN (left) and PNP (right), showing their terminals: Base (B), Collector (C) and Emitter (E) [5].

The main function of a transistor is to control the flow of current, acting as a switch or, alternatively, as an amplifier. When amplifying a signal, the transistor is able to increase the amplitude of a small signal applied to its base, resulting in a higher power signal at the collector [1].

The relation between a transistor and an amplifier is that the transistor has its three terminals that can be designed as amplifiers with specific characteristics of gain, impedance, and frequency [3].

While a transistor is an individual component capable of controlling and amplifying signals, an amplifier is a complete circuit that works with transistors, among other components, in order to increase the amplitude of an input signal according to the desired needs [4].

II. THEORETICAL FRAMEWORK: AUDIO AMPLIFIER BY MEANS OF A VOLTAGE DIVIDER

There are different configurations when amplifying a signal; in this section we discuss an audio amplifier based on a voltage divider, which basically is a circuit that uses this specific configuration to establish the necessary bias level in the transistor, allowing it to operate efficiently in the active region. This configuration is characterized by being simple and reliable, since the voltage divider, formed by two resistors in series, divides the supply source to generate a constant voltage that controls the transistor's behavior [2].

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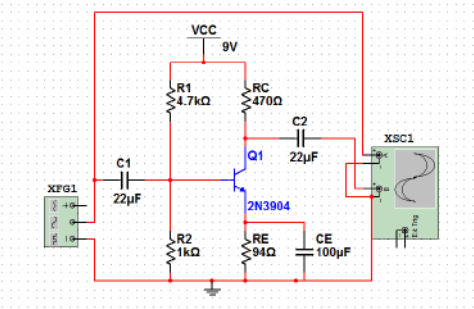


Fig. 2: Schematic of an audio amplifier based on a voltage divider using a 2N3904 transistor. Includes resistors for biasing (R1 and R2), collector resistor (RC), emitter resistor (RE), a decoupling capacitor (CE) and coupling capacitors (C1 and C2) for input and output signal, powered by a 9V source [6].

Figure 2 shows the schematic of an audio amplifier based on a voltage divider using an NPN transistor model 2N3904. The circuit uses resistors for biasing, coupling and decoupling components, and a 9V power supply.

A voltage divider is typically composed of two resistors in series connected between the supply voltage (VCC) and ground. The voltage at the junction of these resistors is used to bias the transistor's base. If R1 and R2 are the divider resistors, the bias voltage is given by:

$$VB = \frac{R2}{R1 + R2} VCC \quad (1)$$

This voltage determines the base current (IB) of the transistor, which in turn controls the collector current (IC) according to the relation:

$$IC = \beta IB \quad (2)$$

where β is the current gain of the transistor.

A. Components

The circuit also includes additional resistors and capacitors to improve its performance:

- RE and RC: The RE resistor in the emitter helps stabilize the operating point, while RC determines the amplifier's gain.
- CE: The bypass capacitor CE removes the alternating current component in RE, increasing the amplifier's gain.
- C1 and C2: The coupling capacitors allow only alternating current signals to pass between the input and output stages, blocking any direct current component.

The frequency response of the amplifier is determined by the capacitors and resistors in the circuit. The lower cutoff frequency (fc) of a capacitor C with a in series resistor R is calculated as:

$$fc = \frac{1}{2\pi RC}. \quad (3)$$

This parameter is crucial to ensure that the amplifier operates in the frequency range (20 Hz - 20 kHz).

III. METHODOLOGY

To carry out this project, a structured but practical process was followed [7], which covers from the initial idea to the validation of the final design. Below, the stages carried out are described.

A. Circuit design

For the circuit design of an audio amplifier based on transistors, a biasing network using a voltage divider served as the basis. To achieve more efficient results in the project, two amplifiers were built, meaning the circuit design was divided into two phases.

1) First stage of amplification: In the first stage of amplification, the circuit sketch was made (Fig. 3.) and afterwards the relevant calculations. In this phase, the guitar input is present, followed by the connection of a transistor in voltage division with capacitors.

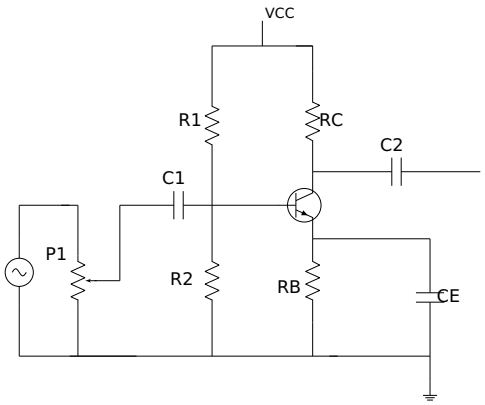


Fig. 3: Initial schematic of the first amplification stage. This preliminary circuit defines the basic configuration for signal amplification, including the use of a transistor as the active element and passive components such as resistors and capacitors to stabilize and control the gain. The values of the components, such as R1, R2, RE, C1, C2, and CE, have not yet been determined, allowing for later adjustment to meet design specifications.

In the first amplification stage, a 2N2222A transistor is used as a preamplifier, which is expected to achieve a collector current gain of 100. For this, suitable resistors were chosen to ensure a collector current of 15 mA and a base current of 150 µA. To select these resistors, the analysis began by setting the collector and emitter pins at 50% of the supply voltage to allow a signal range with minimal distortion, in this case 18V, and the emitter resistor is expected to have 10% of the supply voltage. Therefore, for

determine the value of the required resistance, the following calculation was carried out.

$$RE = \frac{10\% \times 18V}{15mA} = 120\Omega \quad (4)$$

For the collector resistor it was assumed that the current in the collector and emitter is the same (IE = IC); to obtain 50% of the supply voltage across the emitter-collector, the collector resistor must drop 40% of the supply voltage, making a calculation similar to the previous one, we obtain:

$$RC = \frac{40\% \times 18V}{15mA} = 480\Omega \quad (5)$$

For the resistors of the voltage divider located at the base of the transistor we ensured that R2is 10 times larger than the emitter one, guaranteeing a base current smaller than that in the collector and emitter.

$$R2 = 10 \times Re = 1.2K\Omega \quad (6)$$

And for the last resistor, the analysis of the voltage that will drop on the base of the transistor was performed; this voltage is known by Kirchhoff's laws, since the voltage at the emitter pin of the transistor was chosen to be 10% of our voltage source, and the base-emitter voltage drop is approximately 0.7V; with this voltage we can calculate the expected voltage drop on the first resistor and thus calculate it with the voltage relation between base, the expected drop, and R2.

$$Vbe \approx 0.7V, Ve = 1.8V \quad (7)$$

$$Vb = 1.8 + 0.7 = 2.5V \quad (8)$$

$$VR1 = 18 - 2.5 = 15.5V \quad (9)$$

$$R1 = \frac{15.5}{2.5} \times R2 = 10.33K\Omega \quad (10)$$

Finally we calculate the power each resistor will dissipate to ensure we buy the right type.

$$PRE = 1.8V \times 15mA = 27mW \quad (11)$$

$$PRC = 7.2V \times 15mA = 108mW \quad (12)$$

$$PR2 = \frac{2.5V2}{1200\Omega} = 5.2mW \quad (13)$$

$$PR1 = \frac{15.5V}{10.333K\Omega} = 23.25mW \quad (14)$$

2) Second amplification stage: For the amplifier of power, we chose the TIP32-C transistor because it has a higher current capability and a moderate power dissipation, but it can be improved up to 20 times with a heat sink; this transistor was given a collector current of IC = 150mA and its base current was adjusted to IB = 15mA to be connected to the first amplification stage.

The procedure used to determine the component values in the first amplification stage was applied, analogously, to calculate the values corresponding to the components in the second amplification stage. This allowed ensuring continuity in the design and coherence in the circuit performance.

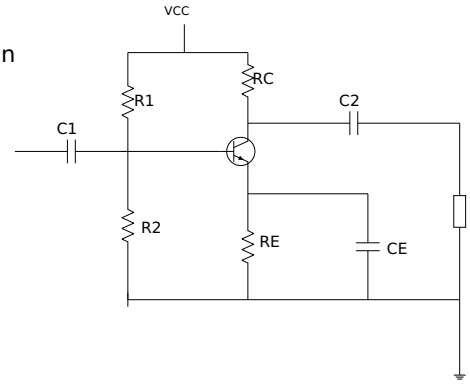


Fig. 4: Schematic of the final amplification circuit. This design shows the complete configuration of the second amplification stage

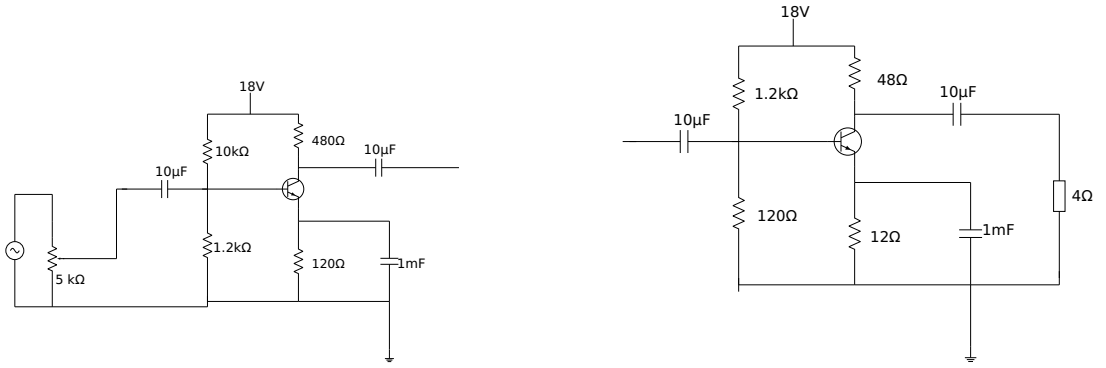
amplifier, where R1 and R2 form a voltage divider to bias the transistor. Capacitors C1 and C2actuate as coupling for the input and output signal, respectively- while CE provides a higher gain by decoupling the RE resistor. The circuit is designed to provide a stable and efficient amplification.

3) Final circuit design: The final circuit design (Fig. 5) was the result of a tuning process based on the previous amplification stages. The main objective was to optimize the total system gain and ensure the circuit was stable and efficient. This design incorporates carefully selected components to meet the specific amplification needs.

B. Circuit Simulation

The circuit simulation in its final stage was performed using the free Tinkercad software, a very useful tool for designing and simulating electronic circuits in a visual and interactive way. The implemented design includes a 2N2222A transistor and a TIP32C, which fulfill the fundamental role for audio amplification, chosen because their gain characteristics and power handling capacity (Fig. 6.).

The circuit configuration includes the voltage divider mentioned earlier, formed by resistors at the base of the 2N2222A transistor, as this sets the proper bias point for the transistor to operate



(a) Circuit 1 (b) Circuit 2

Fig. 5: Amplification circuit in two stages. (a) First stage of amplification: the initial circuit design is presented, where the component values, such as resistors and capacitors, were previously calculated to ensure an efficient increase of the initial signal. This stage establishes the basic gain of the system. (b) Second stage of amplification: shows the final circuit design, where the calculated component values optimize the total gain and stability of the system. This stage reinforces the amplification achieved in the first, ensuring an output signal suitable for the desired purpose.

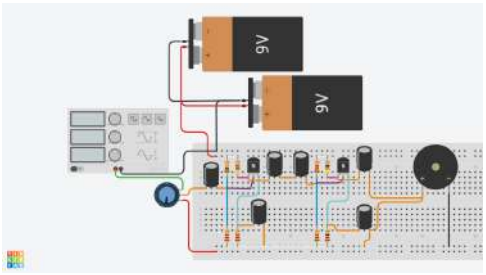


Fig. 6: Visualization of the simulation of the final stage of the amplifier circuit, made in Tinkercad.



Fig. 7: The transistor in the image has the same package (TO220) as the TIP32C used in the circuit design [8].

appropriately. On the other hand, coupling capacitors (C1 and C2) were used to ensure a clean transfer of the alternating current signal between each layer, thus removing the undesired direct current from other components. Additionally, a decoupling capacitor (CE) was integrated, which helps to eliminate the voltage drop in the emitter resistor, thus maximizing gain.

For the power supply, it was determined to use 18V, so two 9V batteries were chosen, which provide the energy needed for the circuit to operate.

C. Construction of the circuit

After simulating the circuit and determining the optimal values for each component, it was assembled on the breadboard, subsequently performing the tests and adjustments necessary to ensure proper operation.

IV. RESULTS

With the construction of the audio amplifier, practical tests were carried out to evaluate its behavior under real conditions. The circuit was connected to a speaker

Radox 065-176 of 5 inches with a 15W rating and an impedance of 4Ω, reproducing audio signals from an electric guitar. The results showed that it successfully amplifies the audio signal, allowing its output at a perceptible volume. However, distortion is noticeable at the output. Additionally, an increase in the temperature of the transistors and some resistors was observed during prolonged operation. Even with the use of a heat sink, it is not enough to keep the temperature steady.

in a range optimal, evidencing the need for improvements in the thermal management of the circuit. This high thermal dissipation affects the efficiency and durability of the amplifier, requiring the use of more powerful heat sinks. In general, the amplifier meets its function, but it presents aspects that require improvement in output quality and performance during prolonged use without heating up as much.

V. ANALYSIS OF RESULTS

The built audio amplifier showed acceptable performance, capable of amplifying audio signals from an electric guitar. However, the results analysis led us to identify areas where improvements are needed for much more optimal and reliable performance.

For example, although the amplification was sufficient to reproduce a good sound volume, harmonic distortion was detected. Also, during prolonged use, the TIP32C transistor showed heating, along with the closest resistors even with a heat sink, and this could significantly reduce the useful life of the component and the proper functioning of the amplifier. Or the robustness of the design, which although allowed us to validate the functionality of the circuit, the protoboard is not so practical; ideally, it would be a small and compact design on a PCB, which would have much more stable connections and avoid noise or false contacts. In addition, during the construction stage while selecting the components, it was not possible to obtain

resistors of 48Ω exact ones, so two connected in resistors of 100Ω parallel were used to obtain an approximate value to the 48Ω. Similarly, for 12Ω resistors it was necessary to use some of 24Ω in parallel. These are modifications that slightly affect the behavior of the circuit and should be improved. In short, although the objectives were met, improving all these aspects will make the amplifier more efficient and stable, suitable for more demanding uses.

VI. CONCLUSION

The design, simulation, and construction of the audio amplifier based on transistors managed to fulfill its main objective of amplifying audio signals for playback at an adequate volume in output devices, such as speakers. The implementation of the amplification stages, using the 2N2222A and TIP32C transistors, allowed for correct gain and satisfactory performance. However, during the analysis of results, it was determined which areas need to be improved for better amplifier performance. During testing, heating of the TIP32C transistor was observed, even when integrating a heat sink, which compromises the thermal efficiency of the circuit. Likewise, the quality of the output signal showed slight distortions. These issues are significant for modifying the design for better thermal resistance. Another important aspect was the selection of components, specifically the resistors. Not having used the exact resistors that were calculated due to complications related to availability in stores led to the use of parallel combinations, which has slight repercussions on the design's precision. Finally, although the use of the protoboard is practical for checking the amplifier's functionality, for more reliable and durable use, it was determined that ideally it should be transferred to a PCB to make it more compact and with more precise connections to avoid false contacts. In conclusion, the development of this audio amplifier based on transistors enabled the exploration of both the fundamental principles of amplification and the practical challenges that arise in the implementation of electronics. Although the project achieved satisfactory basic functionality, it showed several technical aspects that require improvements to increase the system's performance.

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